

Human vs. AI in Symbolic Music: An Online Listening Study on Bach-Style Chorales

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Introduction & Method

We conducted a music perception study to evaluate human vs. AI discrimination in Bach-style chorales. Participants ($n = 295$) listened to **20 isolated musical phrases** and categorized each as **AI-** or **human-composed**. AI snippets were generated using *NotaGen* [1], a state-of-the-art symbolic music generation model. All musical snippets (both human and AI) were manually curated. Participation was open to the public with **no prior music qualification required**. Prior to and following the experiment, we collected metadata such as age, musical background, and self-reported confidence levels in order to better contextualize participant performance.

Experiment Design

Pre-Survey → **20 Listening Trials** → **Post-Survey**
(Order randomized)

Stimuli

- ▶ **Balanced Set:** 10 Bach chorale phrases vs. 10 *NotaGen* phrases
- ▶ **Audio Uniformity:** All excerpts rendered with an identical MIDI harpsichord soundfont
- ▶ **Selection:** All snippets were manually curated

Measured Parameters

- ▶ **Demographics:** Age, weekly AI usage (Yes/No).
- ▶ **Musical Background:** Plays instrument (Yes/No), sings in choir (Yes/No), self-reported experience (4-point Likert scale).
- ▶ **Confidence:** Pre- and post-experiment self-ratings (4-point Likert scale).

Overview and Results

Participant Overview

- ▶ **Sample Size:** $n = 295$ participants (primarily German, aged 15–86).
- ▶ **Musical Background:** $\approx 90\%$ play an instrument; $> 50\%$ sing in a choir.
- ▶ **AI Usage:** $\approx 50\%$ use AI multiple times per week.

Participant Feedback¹

- ▶ "frighteningly difficult 😬"
- ▶ "It was enormously difficult to hear a difference. I was constantly guessing. If I guessed something correctly, it merely was a gut instinct. [...]"
- ▶ "It was a very interesting experience because I realized, how damn hard it is to me distinguish AI compositions from human compositions."
- ▶ "The snippets are relatively short. Longer snippets could create a higher hit rate, I guess..."
- ▶ "It's really exciting because it's really difficult to decide whether it was AI or human after hearing it only once. Sometimes I would have needed it a second time."

¹Quotes were translated from German to English

Sample Skews Young

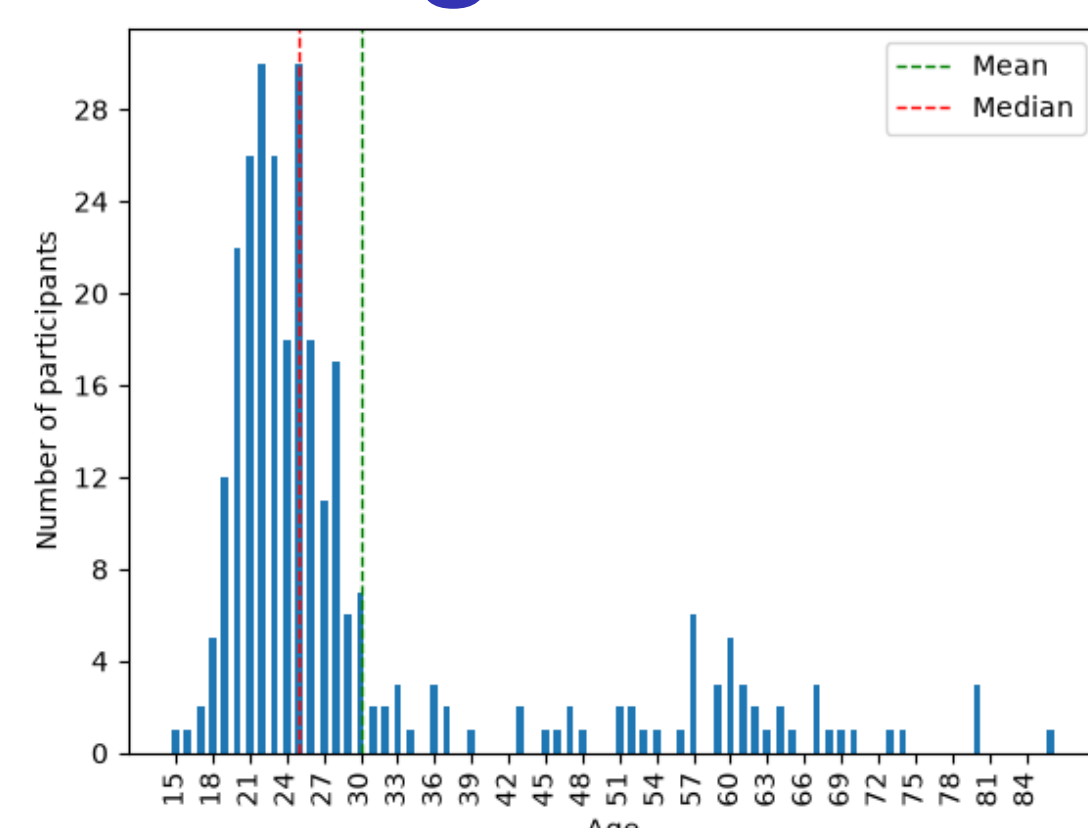


Figure 1: Participant age distribution ($n = 295$).

Accuracy By Musical Experience

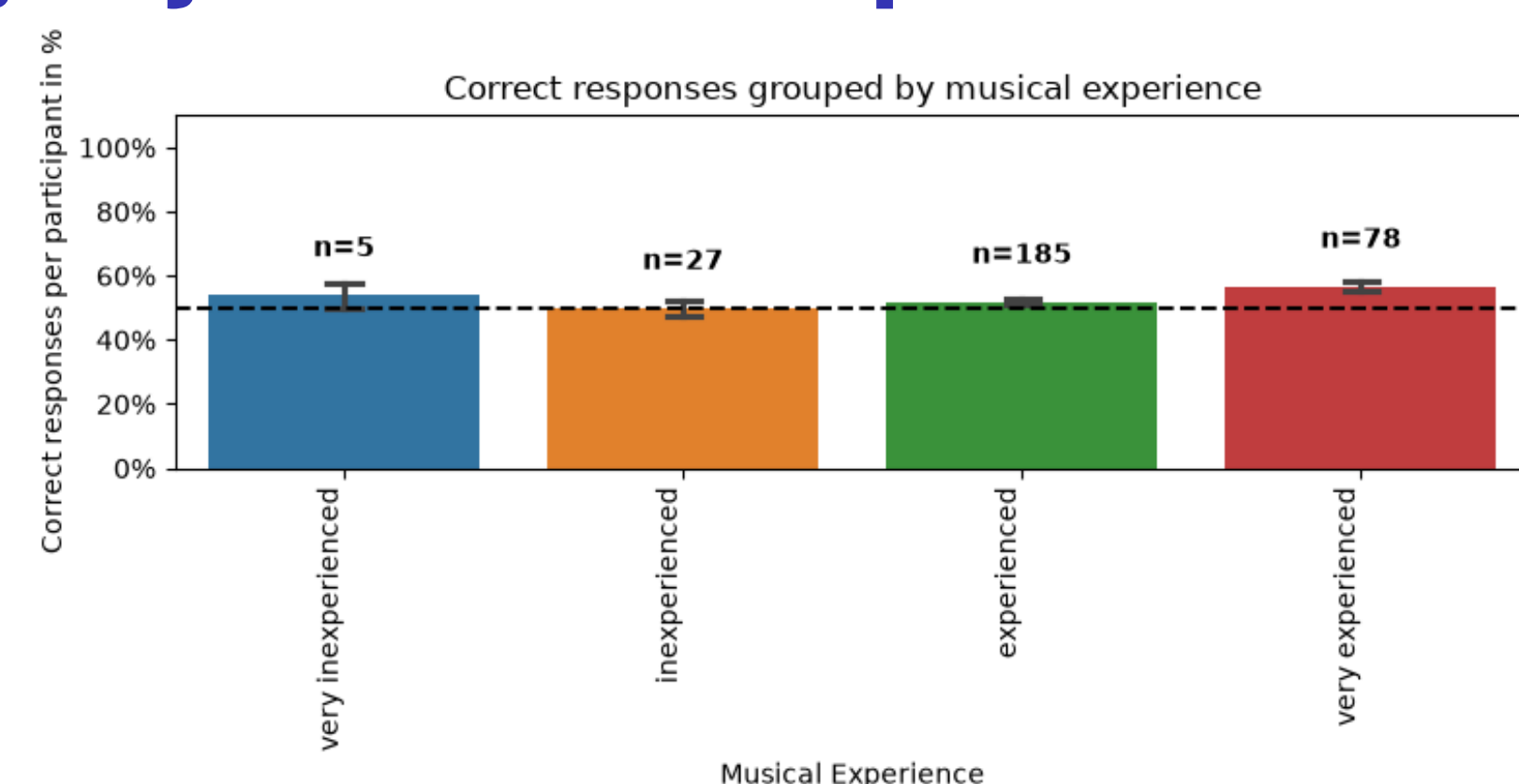


Figure 2: Discrimination accuracy across participants' prior musical experience.

Choir Singers Outperform Non-Singers

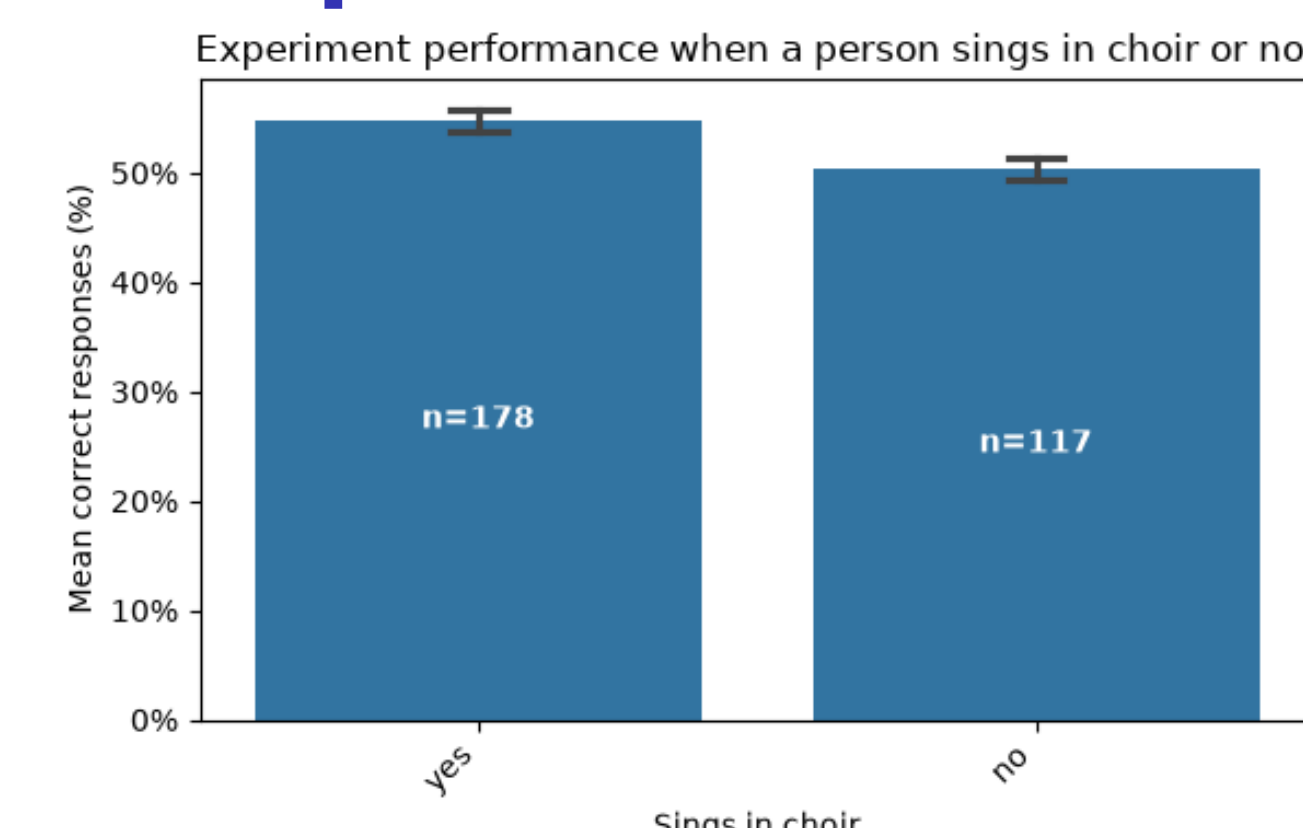


Figure 3: Accuracy comparison by participants' choir experience.

Trial Exposure Shifts Confidence

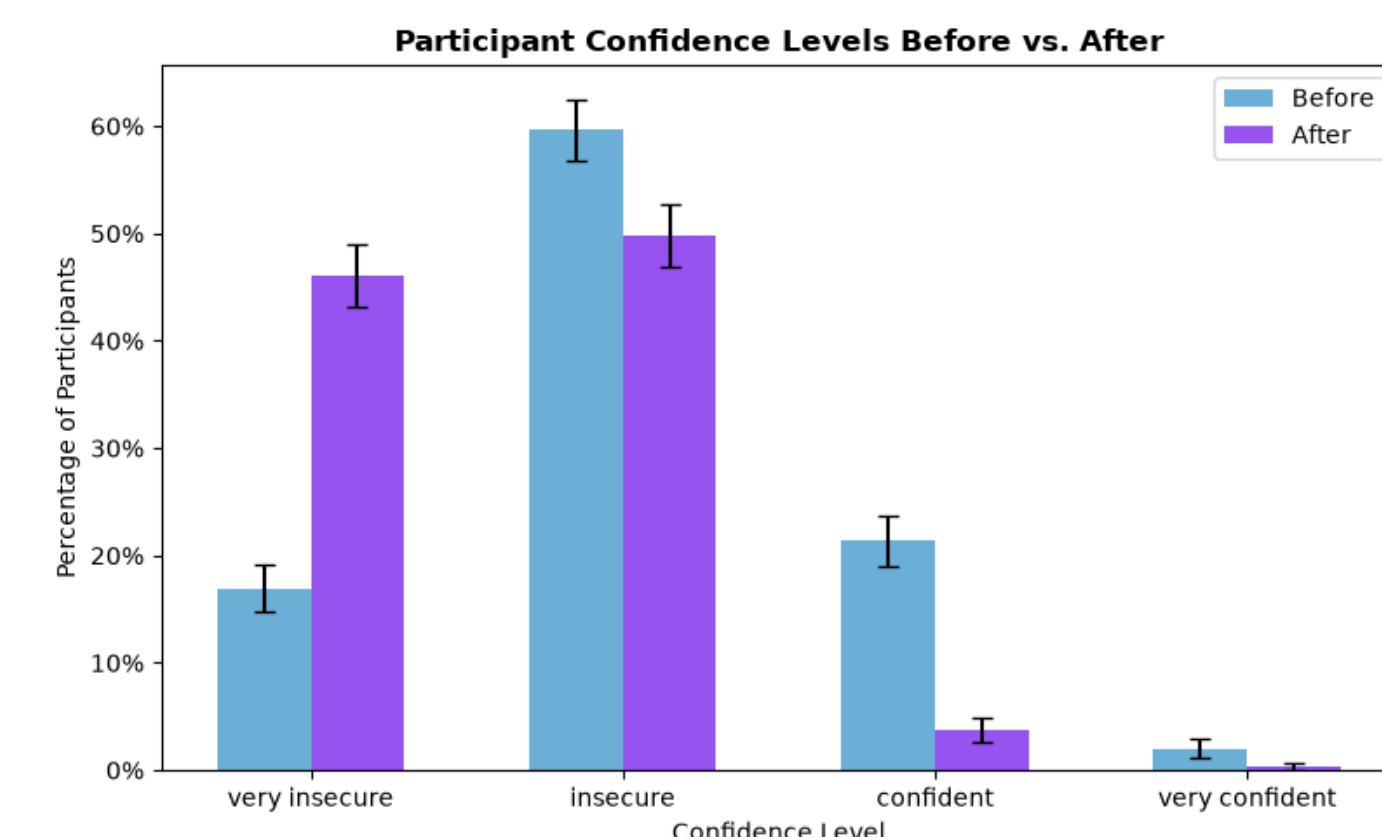


Figure 4: Self-reported confidence pre- vs. post-trial.

Overview: Performance Per Participant

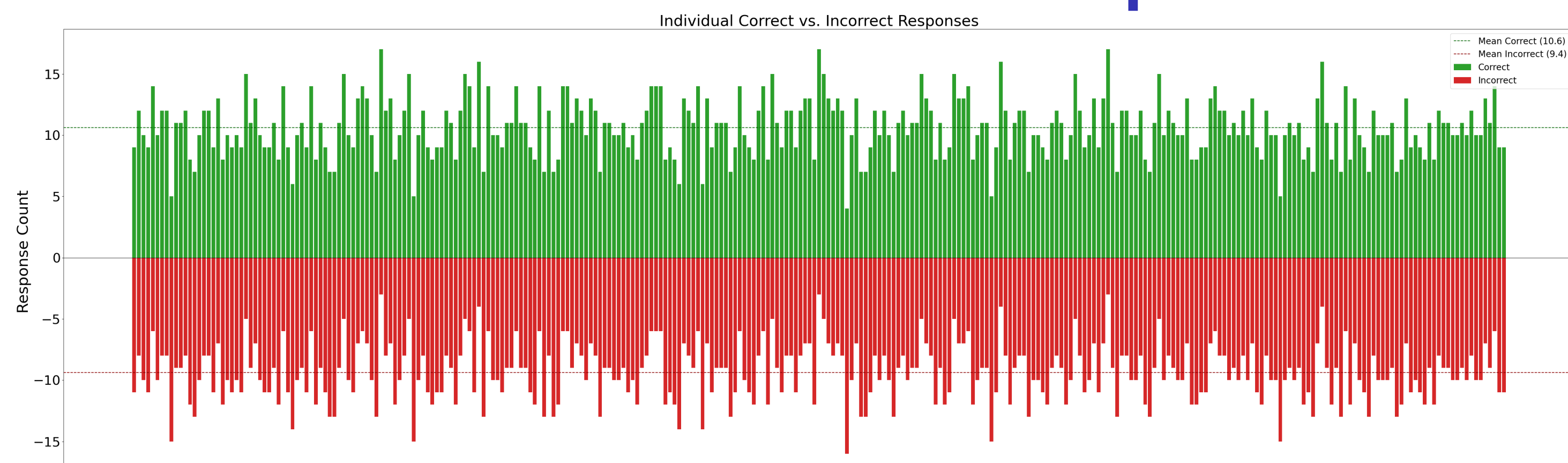


Figure 5: Individual participant accuracy scores across all trials.

Stimulus Difficulty Varies

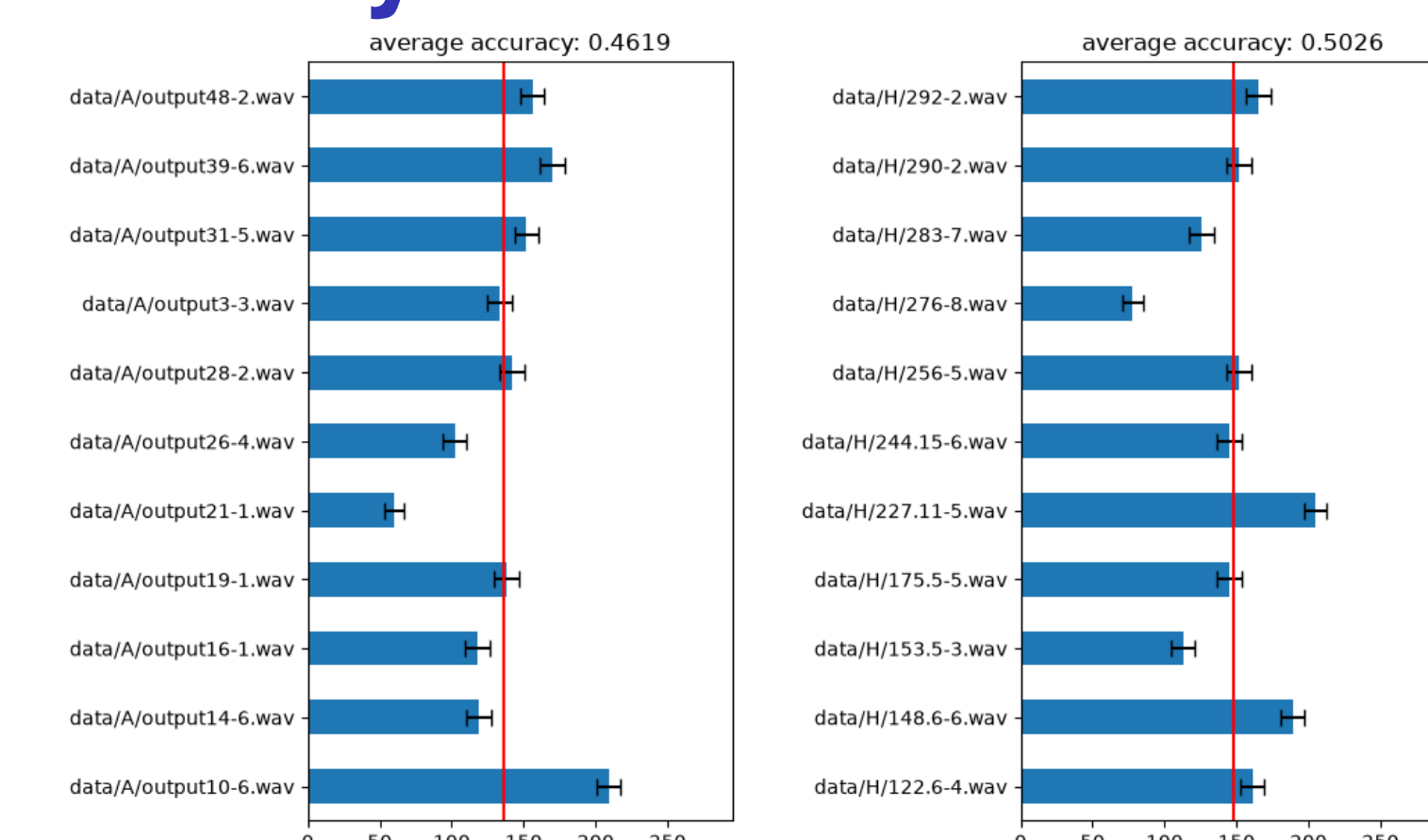


Figure 6: Overview of individual stimulus discrimination. Left plot shows AI snippets. Right plot shows human-composed snippets.

Limitations and Future Work

- ▶ MIDI playback may lead to artifact bias.
- ▶ Potential selection bias in chosen stimuli.
- ▶ Lack of visual sheet music representation.

- ▶ Comparison with other symbolic music generation models (AI vs. AI / AI vs. Human)
- ▶ Implement longer musical stimuli
- ▶ Expand survey towards more "artistic" musical genres

References

- [1] Yashan Wang et al. *NotaGen: Advancing Musicality in Symbolic Music Generation with Large Language Model Training Paradigms*. Mar. 2025. DOI: 10.48550/arXiv.2502.18008. arXiv: 2502.18008 [cs]. URL: <http://arxiv.org/abs/2502.18008> (visited on 01/21/2026).

Try the Experiment



Can you distinguish AI from human-composed?

← Scan to test your skills

jasonvla.github.io/ai-chorales_website